

MOHIT PRAKASH MOHANTY

Associate Professor, Department of Water Resources Development and Management,
Indian Institute of Technology Roorkee, Roorkee, Uttarakhand, India
Schmidt AI in Science Faculty Fellow, University of Oxford, UK



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PRESENT STATUS

Associate Professor *June 2026 to present*
Department of Water Resources Development and Management
Indian Institute of Technology Roorkee

RESEARCH EXPERIENCE AND EDUCATION

Assistant Professor *May 2021 to June 2026*
Department of Water Resources Development and Management
Indian Institute of Technology Roorkee

Post-Doctoral Fellow

Oct. 2019 to Apr. 2021
The University of Western Ontario, Canada
Host Professor: [Prof. Slobodan P. Simonovic](#)
Project title: Large-Scale Floodplain Mapping using various Publicly Available Datasets
Funding organizations- Natural Sciences and Engineering Research Council (NSERC)

Research Associate

June 2019 to Aug. 2019
Indian Institute of Technology Bombay, India
Project title: Flood Hazard Mapping of Data-scarce
Large Flood-prone Regions using Geomorphic Classifiers
Supervisor(s): [Prof. Subhankar Karmakar](#)

Doctor of Philosophy (Ph.D.)

Jan. 2015 to Aug. 2019
[Recipient of the "Naik and Rastogi Excellence in Ph.D. Thesis"-
<https://www.iitb.ac.in/newacadhome/2021ThesisAwardies.html>]
Indian Institute of Technology Bombay, India
Thesis title: A comprehensive mapping of Flood Risk in a Changing Climate:
An application to Jagatsinghpur District, Odisha
Supervisor(s): [Prof. Subhankar Karmakar](#) and [Prof. Subimal Ghosh](#)

Master of Technology (M.Tech. in Environmental Engineering)

Jul. 2012 to Jul. 2014
Indian Institute of Technology Guwahati, India
Supervisor: [Prof. Pranab K. Ghosh](#)
Thesis title: Studies on the removal of nitrate and phenol in the
presence of sulphate using a mixed microbial consortium

Bachelor of Technology (B.Tech.)

Aug. 2007 to Jul. 2011
Department of Civil Engineering
Institute of Technical Education and Research
Siksha 'O' Anusandhan University, India

RESEARCH INTERESTS

Multi-Hazard Risk Management: Compound extremes, Hydrodynamic flood inundation modeling, and hazard mapping, Geomorphic classifiers, Socio-economic and infrastructure vulnerability analysis, Glacial Lake Outburst Floods (GLOFs), Hydrological modeling, Drought risk management, Climate change resilience, and adaptation, Socio-Hydrology, AI-ML techniques for hydrological predictions, Remote Sensing and GIS

Climate Resilience and Adaptation: Climate-resilient strategies for water resources and infrastructure, Adaptation frameworks for community and ecological resilience

Hydro-environmental modeling: Sustainable Drainage Systems (SuDS), Lake Rejuvenation, Water, Sanitation, and Hygiene (WaSH), Human-health risks from hydro-meteorological extremes

PROFESSIONAL CONTRIBUTIONS

Editorial Contributions

- **Associate Editor**, *Journal of Hydrology, Elsevier* (Q1, Impact Factor: 6.3)
- **Associate Editor**, *The Journal of Flood Risk and Management, Wiley* (Q1, Impact Factor: 3.0)
- **Associate Editor**, *Journal of Water and Climate Change, IWA Publishing* (Q2, Impact Factor: 3.1)
- **Editor**, *Scientific Reports, Nature Portfolio* (Q1, Impact Factor: 3.8)
- **Editor**, *Scientific Review, Nature Portfolio*
- **Editor**, *Environmental Science and Technology (ES&T) Water, American Chemical Society (ACS) Publications* (Q1, Impact Factor: 4.8)
- **Editor**, *Discover Hazards, Springer*
- **Editor**, *Journal of the Indian Water Resources Society (IWRS)*
- **Guest Editor**, Special Issue on “Advancing Risk Assessment and Resilience Strategies for Climate Extremes and Multi-Hazard Events” in the *Environmental Monitoring and Assessment, Springer* (Q2, Impact Factor: 3.6) [*Ongoing*]
- **Guest Editor**, Special Issue on “Agricultural Water Management for a Sustainable Future: Technologies and Strategies” in the *Environmental Monitoring and Assessment, Springer* (Q2, Impact Factor: 3.6)
- **Lead Guest Editor**, Special Issue on “Sustainable Technologies to Quantify Risks from Climate and Weather-related Disasters” in the *Environmental Monitoring and Assessment, Springer* (Q2, Impact Factor: 3.6)

EXPERT REVIEW COMMITTEE

- Expert Reviewer for the “*Special Report on Climate Change and Cities*”, developed under the **Intergovernmental Panel on Climate Change (IPCC) AR7 cycle**
- Member of the **Bureau of Indian Standards (BIS) Sectional Committee** to design standards on *urban flood management*
- Member, **Task Force on rejuvenation of lakes** in Goa

MEMBERSHIPS

- **Member, International Scientific Committee**, 10th International Conference on Flood Management to be held in London, Ontario, 20-26 May 2026
- **Member**, Bureau of Indian Standards (BIS), Sectional Committee, Urban Flooding
- **Member, Technical Advisory Committee**, HYDRO 2025: 30th International Conference on Hydraulic, Water Resources, Environmental, and Coastal Engineering, organized by the National Institute of Technology Rourkela, India, during December 18–20, 2025, under the aegis of the Indian Society for Hydraulics (ISH), Pune.

- **Member, Technical Committee**, Innovative Design for Excellence, Affordability, and Sustainability (IDEAS-2024), held at IIT Roorkee from Oct. 19-20, 2024
- **Member**, American Geophysical Union (AGU)
- **Member**, European Geophysical Union (EGU)
- **Member**, Indian Water Resources Society (IWRS)

ACADEMIC HONORS and AWARDS

- **The Eric and Wendy Schmidt AI in Science Faculty Fellowship Programme at Oxford University, 2026**
- **International Travel Grant** to attend the Multi-hazard DRR Academy held in Barcelona, Spain, from Oct. 23-26, 2024
- **Sivapalan Young Scientists Travel Awards (SYSTA)** award from **IAHS (International Association of Hydrological Sciences)** to attend the 9th International Symposium on Integrated Water Resources Management (IWRM) held in Florianópolis, Brazil from 4-7 November 2024
- **International travel grant** from UNESCO for attending the 9th International Conference on Flood Management (ICFM9) held in Tsukuba, Japan, from Feb. 18-22, 2023
- **Naik and Rastogi Award** for Excellence in Ph.D. Research for the year 2019-2021
- **Post-Doctoral Fellowship** at The University of Western Ontario, Canada from Oct. 2019 to Mar. 2021
- **International travel grant** from the National Science Foundation (NSF) for attending the Eighth International Conference on Flood Management held in Iowa, USA, from Aug. 17-19, 2021
- **International travel grant** from the Science and Engineering Research Board (SERB), Department of Science and Technology (DST), Govt. of India for attending the 11th World Congress on Water Resources and Environment held in Spain from Jun. 25-29, 2019
- **International travel grant** from CORDEX and Asia-Pacific Network for Global Research (APN) for attending ICRC CORDEX 2016 held in the University of Stockholm, Stockholm, Sweden, from 17-22 May 2016
- **Ph.D. scholarship** from Space Technology Cell- IIT Bombay Indian Space Research Organisation (Jan. 2015-Aug. 2019).
- The **Graduate Aptitude Test in Engineering (GATE) 2012** scholarship for higher studies
- The second position (in terms of CPI) in undergraduate studies at the university level

LIST OF PUBLICATIONS (*: *Corresponding Author*)

(Total citations: 2004, h-index: 21, i10-index: 33, as of 13 August 2026)

Google Scholar: <https://scholar.google.com/citations?user=7THkWAEEAAA&hl=en>

ORCID: <https://orcid.org/0000-0001-7783-2073>

SCOPUS: <https://www.scopus.com/authid/detail.uri?authorId=57203633749>

Research Gate: <https://www.researchgate.net/profile/Mohit-Mohanty-2>

Publications in refereed journals

- J1. Tripathy, P., Thakur, D. A., & **Mohanty, M.P.*** (2026) From Single to Compound Hazards: How Flood and Drought Vulnerabilities have Intensified across India (npj Natural Hazards, Nature, **Q1**) [*in press*]
- J2. Tripathi, V., AghaKouchak, A., Karmakar, S., Stein, L., and **Mohanty, M.P.*** (2026). Interpretable Deep Learning Reveals Influence and Interactions of Hydroclimatic Drivers on Floods Across India. Earth's Future, AGU Publications, Wiley 14, e2026EF008397. (**Q1, Impact Factor: 8.5**) <https://doi.org/10.1029/2026EF008397>
- J3. Deopa, R., Tripathi, V., Nandana, P. S., & **Mohanty, M. P.*** (2026). Coupled hydrological and public health risks from urban flooding: integrated remote sensing, machine learning, and

- hydrodynamic–ecological modelling. *Journal of Hydrology*, 135999. (Q1, Impact Factor: 7.3) <https://doi.org/10.1016/j.jhydrol.2026.135999>
- J4. Legese, B., Pandey, A.*, and Mohanty, M.P. (2026). Predictive Ecosystem Health Assessment under Land-Use and Vegetation Dynamics in a Semi-Arid Rift Valley Basin, Ethiopia, *Land Degradation & Development*, Wiley (Q1, Impact Factor: 4.0)
- J5. Tyagi, M., Mishra, A., & **Mohanty, M.P.*** (2026). Compound Hydroclimatic Extremes under Anthropogenic Influence: Transition Patterns and Population Exposure Risks. *Earth's Future*, AGU Publications, Wiley (Q1, Impact Factor: 8.5) <https://doi.org/10.1029/2025EF007474>
- J6. Rachit, Tripathi, V., & **Mohanty, M.P.***, Pandey, A., & Gupta, A. (2026). Rapid Flood Susceptibility Mapping in the Indian Himalayan Region using CNN-U-Net Segmentation: Insights from the 2025 Monsoon Events. *Natural Hazards*, Springer (Q1, Impact Factor: 4.5) <https://doi.org/10.1007/s11069-026-08175-w>
- J7. Thakur, D. A., Sharma, A., & **Mohanty, M.P.*** (2026). Development of Multi-Hazard Flood Risk Information System for Coastal Region: A Consolidated End-To-End Web Platform for Efficient Flood Management. *Environmental and Sustainability Indicators*, Elsevier (Q1, Impact Factor: 5.6) <https://doi.org/10.1016/j.indic.2026.101115>
- J8. Sharma, P., Tripathi, V., & **Mohanty, M.P.*** (2026). Improving Daily Streamflow Predictions over Large Watersheds: Introducing a Novel Enhanced Long Short-Term Memory (En-LSTM) model. *Water Resources Management*, Springer (Q1, Impact Factor: 4.7) <https://doi.org/10.1007/s11269-025-04475-1>
- J9. Thakur, D. A., Vishveshwari, T.S., & **Mohanty, M.P.*** (2025) Nuisance Flood Risk: Defining a New Horizon in Urban Flood Risk Management through Hydrodynamic Flood Hazard Modelling and Indicator-Based Vulnerability Assessment. *International Journal of Disaster Risk Reduction*, Elsevier (Q1, Impact Factor- 4.5) <https://doi.org/10.1016/j.ijdrr.2025.105969>.
- J10. Singh, H., & **Mohanty, M.P.*** (2025). Multi-Scale Assessment and Entropy-MCDM Framework for Evaluating Reanalysis Precipitation Datasets over Indian Basins. *International Journal of Applied Earth Observation and Geoinformation*, Elsevier (Q1, Impact Factor: 8.6) <https://doi.org/10.1016/j.jag.2025.104919>
- J11. Legese, B., Pandey, A., * & **Mohanty, M.P.** (2025) A Novel Integrated Framework for Long-Term Assessment of Ecosystem Service Degradation and Restoration Prioritization in a Semi-Arid Rift Valley Landscape. *Environmental Monitoring and Assessment*, Springer (Q2, Impact Factor: 3.6) <https://doi.org/10.1007/s10661-025-14724-w>
- J12. Deopa, R., Singh, H., Tripathi, V., Tyagi, M., & **Mohanty, M.P.*** (2025). Cascading Impacts of 2025 July multi-hazard episodes in the Western Himalayas: Evidence through Remote Sensing observations and Google Earth Engine. *Natural Hazards*, Springer (Q1, Impact Factor: 3.7) <https://doi.org/10.1007/s11069-025-07699-x>
- J13. Deopa, R., & **Mohanty, M. P.*** (2025). HyEco: A Hybrid Hydrodynamic-cum-Ecological Modelling Framework to Demystify Flood and Human-Health Risks over Flood-prone Metropolis. *Water Research X*, Elsevier 100396. (Q1, Impact Factor: 8.2) <https://doi.org/10.1016/j.wroa.2025.100396>
- J14. Mishra, D., Singh, H., Kumar, M., & **Mohanty, M.P.*** (2025) Advanced Hydrological Assessment with SWAT+ under Climate Change: An Integrated Framework over Large Climate-Sensitive and Reservoir-Regulated Watersheds. *Science of the Total Environment*, Elsevier (Q1, Impact Factor: 8.0) <https://doi.org/10.1016/j.scitotenv.2025.180062>
- J15. Alam, G., Tripathi, V., & **Mohanty, M. P.*** (2025). A Novel Framework Embedding Bayesian-optimized Ensemble Machine Learning and Explainable Artificial Intelligence (XAI) to Improve Flood Prediction in Complex Watersheds. *Environmental and Sustainability Indicators*, Elsevier (Q1, Impact Factor: 5.6) <https://doi.org/10.1016/j.indic.2025.100760>
- J16. Singh, H., and **Mohanty, M.P.*** (2025) Performance of ERA5 Products in Capturing Pluvial and Fluvial Inundation: A Statistical-cum-Hydrodynamic Framework for Resource-

Constrained Large Flood-prone Watersheds. *Water Resources Management*, Springer <https://doi.org/10.1007/s11269-025-04257-9> (Q1, Impact Factor: 4.7)

- J17. Parmar, J., **Mohanty, M.P.**, Karmakar, S.* (2025). Need for judicious selection of runoff inputs in a global flood model. *Environmental Research Letters*, IOP Publishing, 20, 024032 <https://doi.org/10.1088/1748-9326/adaa89> (Q1, Impact Factor: 5.8)
- J18. Singh, R., Thakur, D. A., & **Mohanty, M. P.*** (2024). Can Satellite Precipitation Products Comprehend Rainfall Extremes Over Disaster-Sensitive Mountainous Basins? An Exhaustive Inter-comparison and Assessment Over Nepal. *Earth Systems and Environment*, Springer, 1-21. <https://doi.org/10.1007/s41748-024-00557-z> (Q1, Impact Factor: 5.3)
- J19. Tyagi, M., Singh, H., Thakur, D. A., **Mohanty, M.P.*** (2024). Compound Risks of Floods and Droughts over Multi-hazard Catchments: Revealing association through Hydrodynamic-cum-Statistical modelling and Novel Bivariate Risk Classifier. *Science of The Total Environment*, Elsevier, 175882. <https://doi.org/10.1016/j.scitotenv.2024.177689> (Q1, Impact Factor: 8.0)
- J20. Kumari, R., Das, B.S.*, Devi, K., Khuntia, J.R., and **Mohanty, M. P.** (2024). Hydrodynamic Performance of Raceway Pond using k-w and LES Turbulence models. *Journal of Renewable and Sustainable Energy*, AIP Publishing <https://doi.org/10.1063/5.0220538> (Q3, Impact Factor: 1.9)
- J21. Thakur, D. A., & **Mohanty, M. P.*** (2024). Do disparities exist in flood risks during monsoon and post-monsoon seasons? Comprehending diametric behaviors over coastal multi-hazard catchments. *Environmental Research Letters*, IOP Publishing <https://doi.org/10.1088/1748-9326/ad8c6d> (Q1, Impact Factor: 5.6)
- J22. Tripathi, V., Singh, H., & **Mohanty, M. P.*** (2024). Fidelity of machine learning models in capturing flood inundation through geomorphic descriptors over Ganga sub-basin, India. *European Water*, European Water Resources Association (EWRA), 87-88, 1-10
- J23. Deopa, R., Thakur, D. A., Kumar, S., **Mohanty, M. P.***, & Asha, P. (2024). Discerning the dynamics of urbanization-climate change-flood risk nexus in densely populated urban megacities: An appraisal of efficient flood management through spatiotemporal and geostatistical rainfall analysis and hydrodynamic modeling. *Science of The Total Environment*, Elsevier 175882. <https://doi.org/10.1016/j.scitotenv.2024.175882> (Q1, Impact Factor: 8.0)
- J24. Tripathi, V., & **Mohanty, M. P.*** (2024). Can geomorphic flood descriptors coupled with machine learning models enhance in quantifying flood risks over data-scarce catchments? Development of a hybrid framework for the Ganga basin (India). *Environmental Science and Pollution Research*, Springer, 1-20. <https://doi.org/10.1007/s11356-024-33507-3> (Q1, Impact Factor: 5.8)
- J25. Thakur, D. A., & **Mohanty, M. P.*** (2024). Exploring the fidelity of satellite precipitation products in capturing flood risks: A novel framework incorporating hazard and vulnerability dimensions over a sensitive coastal multi-hazard catchment, *Science of The Total Environment*, Elsevier, 170884 <https://doi.org/10.1016/j.scitotenv.2024.170884> (Q1, Impact Factor: 8.0).
- J26. Thakur, D. A., **Mohanty, M. P.***, Mishra, A., and Karmakar, S. (2024). Quantifying flood risks during monsoon and post-monsoon seasons: An integrated framework for resource-constrained coastal regions, *Journal of Hydrology*, Elsevier, 130683 <https://doi.org/10.1016/j.jhydrol.2024.130683> (Q1, Impact Factor- 6.3).
- J27. Thakur, D. A., & **Mohanty, M. P.*** (2023) A Synergistic Approach towards Understanding Flood Risks over Coastal Multi-Hazard Environments: Appraisal of Bivariate Flood Risk Mapping through Flood Hazard, and Socio-Economic-cum-Physical Vulnerability Dimensions. *Science of The Total Environment*, Elsevier, 901, 166423 <https://doi.org/10.1016/j.scitotenv.2023.166423> (Q1, Impact Factor: 8.0).
- J28. Singh, H., & **Mohanty, M. P.*** (2023). Can atmospheric reanalysis datasets reproduce flood inundation at regional scales? A systematic analysis with ERA5 over Mahanadi River Basin, India. *Environmental Monitoring and Assessment*, Springer, 195(10), 1143 <https://doi.org/10.1007/s10661-023-11798-2> (Q2, Impact Factor: 3.0).

- J29. Namgyal, T., Thakur, D. A., Rishi, D. S., & **Mohanty, M. P.*** (2023). Are open-source hydrodynamic models efficient in quantifying flood risks over mountainous terrains? An exhaustive analysis of the Hindu-Kush-Himalayan region. *Science of The Total Environment*, Elsevier, 165357 <https://doi.org/10.1016/j.scitotenv.2023.165357> (Q1, Impact Factor: 8.0).
- J30. Rawat, A., Panigrahi, N., Yadav, B.*, Jadav, K., **Mohanty, M. P.**, Khouakhi, A., & Knox, J. W. (2023). Scaling Up Indigenous Rainwater Harvesting: A Preliminary Assessment in Rajasthan, India. *Water*, MDPI, 15(11), 2042 <https://doi.org/10.3390/w15112042> (Q1, Impact Factor: 3.0).
- J31. Deroliya, P., Ghosh, M., **Mohanty, M. P.**, Ghosh, S., Rao, K. D., & Karmakar, S.* (2022). A novel flood risk mapping approach with machine learning considering geomorphic and socio-economic vulnerability dimensions. *Science of The Total Environment*, Elsevier, 851, 158002 <https://doi.org/10.1016/j.scitotenv.2022.158002> (Impact Factor- 8.2)
- J32. **Mohanty, M. P.***, & Simonovic, S. P. (2022). A comprehensive approach for floodplain mapping through identification of hazards using publicly available data sets over Canada. *Water*, MDPI, 4(14), 2280 <https://doi.org/10.3390/w14142280> (Q1, Impact Factor- 3.0)
- J33. **Mohanty, M.P.**, Karmakar, S.*, and Durga Rao, K.V.H. (2021). An integrated approach of flood risk assessment over a severely flood-prone coastal region using geomorphic classifiers and socio-economic indicators. *Journal of Geomatics, Indian Society of Geomatics*, 15(2) https://isgindia.org/wp-content/uploads/2021/11/JoG_Special_Oct_2021.pdf.
- J34. **Mohanty, M.P.***, and Simonovic, S. (2021) Changes in floodplain regimes over Canada due to climate change impacts: observations from CMIP6 models. *Science of the Total Environment*, Elsevier, 792, 148323 <https://doi.org/10.1016/j.scitotenv.2021.148323> (Q1, Impact Factor- 8.0).
- J35. **Mohanty, M.P.**, and Karmakar, S.* (2021) WebFRIS: an efficient web-based decision support tool to disseminate end-to-end risk information for flood management, *Journal of Environmental Management*, Elsevier, 288, 112456. <https://doi.org/10.1016/j.jenvman.2021.112456> (Q1, Impact Factor- 8.4).
- J36. **Mohanty, M.P.**, Karmakar, S.* & Ghosh, S. (2021) WEBFRIS: a web-based flood information system for enhanced resilience through risk awareness. *European Water*, European Water Resources Association. 69-70, 23-21.
- J37. **Mohanty, M.P.**, & Karmakar, S.* (2021) Hydrodynamic flood modelling of large regions under data-poor situations - A case study of Jagatsinghpur district, Odisha. *International Journal of Business Analytics* (Special Issue on Data Analytics in Humanitarian Supply Chain and Disaster Relief Operations Under Risk and Uncertainty), 8(2) <https://doi.org/10.4018/IJBAN.2021040101>.
- J38. Satpathy, S., Bhatia, U., **Mohanty, M.P.**, Karmakar, S., & Ghosh, S.* (2021) Flood Evacuation during Pandemic: A multi-objective Framework to Handle Compound Hazard. *Environmental Research Letters*, IOP Publishing <https://doi.org/10.1088/1748-9326/abda70> (Q1, Impact Factor- 5.6). [received the Top-cited paper award, 2024]
- J39. **Mohanty, M.P.*** and Simonovic, S. (2020) Fidelity of reanalysis datasets in floodplain mapping: investigating performance at inundation level over large regions. *Journal of Hydrology*, Elsevier, 125757 <https://doi.org/10.1016/j.jhydrol.2020.125757> (Q1, Impact Factor- 6.3).
- J40. **Mohanty, M.P.*** and Simonovic, S. (2020) Understanding dynamics of flood exposure in Canada with multiple high-resolution population data. *Science of the Total Environment*, Elsevier, 143559 <https://doi.org/10.1016/j.scitotenv.2020.143559> (Q1, Impact Factor- 8.0).
- J41. **Mohanty, M.P.**, S, Nithya., Indu, J., Ghosh, S. & Karmakar, S.* (2020) Sensitivity of various topographic data in flood management: Implications on inundation mapping over large data-scarce regions. *Journal of Hydrology*, Elsevier <https://doi.org/10.1016/j.jhydrol.2020.125523> (Q1, Impact Factor- 6.3).
- J42. **Mohanty, M.P.**, Sherly, M. A., Ghosh, S., & Karmakar, S.* (2020) Tide-Rainfall Flood Quotient: An incisive measure of comprehending a region's response to storm-tide and pluvial flooding. *Environmental Research Letters*, IOP Publishing, 15 (6), 064029. <https://doi.org/10.1088/1748-9326/ab8092> (Q1, Impact Factor- 5.6).

- J43. **Mohanty, M.P.**, Vittal, Yadav, V., Ghosh, S. & Karmakar, S.* (2020) A new bivariate risk classifier for flood management considering hazard and socio-economic dimensions. *Journal of Environmental Management*, Elsevier, 255, 109733 <https://doi.org/10.1016/j.jenvman.2019.109733> (Q1, Impact Factor- 8.4).
- J44. **Mohanty, M.P.**, Sahil, M., & Karmakar, S.* (2020) Flood management in India: A focussed review on the current status and future challenges. *International Journal of Disaster Risk Reduction*, Elsevier, 101660 <https://doi.org/10.1016/j.ijdrr.2020.101660> (Q1, Impact Factor- 4.5).
- J45. Gusain, A., **Mohanty, M.P.**, Ghosh, S., & Karmakar, S.* (2020) Capturing Transformation of Flood Hazard over a Large River Basin under Changing Climate using a Top-down Approach. *Science of the Total Environment*, Elsevier, 138600 <https://doi.org/10.1016/j.scitotenv.2020.138600> (Q1, Impact Factor- 8.0).
- J46. Ghosh, M., **Mohanty, M.P.**, Kishore, P., Karmakar, S.* (2020) Performance Evaluation of potential inland flood management options through a 3-way coupled hydrodynamic modelling for a coastal urban catchment. *Hydrology Research*, IWA Publishing, nh2020123 <https://doi.org/10.2166/nh.2020.123> (Q2, Impact Factor- 2.7).
- J47. Ghosh, S.*, Karmakar, S., Saha, A., **Mohanty, M.P.** et al. (2019). Development of India's first integrated expert urban flood forecasting system for Chennai. *Current Science*, 117(5), 741-745 (Q3, Impact Factor- 1.169).
- J48. Brahmacharimayum, B., **Mohanty, M.P.**, & Ghosh, P.K.* (2019). Theoretical and Practical Aspects of Biological sulfate Reduction: A Review. *Global NEST*, 21 (2), 222-244 <https://doi.org/10.30955/gnj.002577> (Q3, Impact Factor- 1.2).
- J49. **Mohanty, M.P.**, Karmakar, S.*, & Ghosh, S. (2018). Regionalized Design Rainfall Estimation: an appraisal of Inundation Mapping for Flood Management under Data-Scarce Situations. *Water Resources Management*, Springer, 32(14), 4725-4746 <https://doi.org/10.1007/s11269-018-2080-8> (Q1, Impact Factor- 4.7).
- J50. **Mohanty, M.P.**, Brahmacharimayum, B., & Ghosh, P. K.* (2018). Effects of phenol on sulfate reduction by mixed microbial culture: kinetics and bio-kinetics analysis. *Water Science and Technology*, IWA Publishing, 77(4), 1079-1088 <https://doi.org/10.2166/wst.2017.630> (Q2, Impact Factor- 2.6).

Manuscripts under an advanced stage of review (*: Corresponding Author)

- J1. Regime-dependent performance of baseflow separation methods across monsoon-dominated catchments in India. (*Environmental Research Letters*, IOP Publishing, Q1, Impact Factor: 5.6)
- J2. Dissanayake, D.M.K.P., Tyagi, M., Wickramaarachchi, T.N.*, and **Mohanty, M. P.*** (20XX) Spatiotemporal Evolution of Meteorological and Hydrological Droughts under Climate Change: A Process-Based Assessment over Kirindi Oya Basin, Sri Lanka (*Journal of Hydrology: Regional Studies*, Elsevier, Q1, Impact Factor: 5.6)
- J3. Saha, S., Singh, H., Sharma, N., & **Mohanty, M.P.*** (202X) Multi-Scenario GLOF Inundation in Nepal: Insights from Three Decades of Remote Sensing and Global Hydrodynamic Modelling (*Physics and Chemistry of the Earth, Parts A/B/C*, Elsevier, Q1, Impact Factor: 4.6)
- J4. Tyagi, M., & **Mohanty, M.P.*** (202X) Deciphering wet-dry event pathways and land-use controls across India (*Environmental Research Letters*, IOP Publishing, Q1, Impact Factor: 5.6)
- J5. Patra, N. R., Thakur, D.A., **Mohanty, M.P.**, & Karmakar, S.* (20XX). A Unified Multiscale Framework for Flood Hazard and Resilience Assessment: Step Towards Efficient Coastal Flood Management (*Environmental Research Communications*, IOP Publishing, Q1, Impact Factor: 2.7)

CONFERENCE PROCEEDINGS (*: Corresponding Author)

- C1. Tripathi, V., & **Mohanty, M.P.*** (2026) Ungauged Tributary Discharge Reconstruction for Monsoon Flood Inundation Modelling: A Geomorphic-Recession Approach Applied to Peninsular India, EGU General Assembly 2026, May 3–8, 2026, Vienna, Austria.
- C2. Saha, S., Singh, H., Sharma, N., **Mohanty, M. P.*** (2026) Nationwide Multi-Scenario GLOF Hazard Mapping in Nepal Using Remote Sensing and Hydrodynamic Modelling, EGU General Assembly 2026, May 3–8, 2026, Vienna, Austria.
- C3. Deopa, R., Debasish, M., Dayal, D., Shahi, N., & **Mohanty, M. P.*** (2026). Hydrology–Health Nexus in a Changing Climate: Multi-Hazard Modelling of Cascading Flood–Health Risks in Urban Megacities. EGU General Assembly 2026, May 3–8, 2026, Vienna, Austria. (*received a student travel grant to present*)
- C4. Rachit, **Mohanty, M. P.***, Pandey, A. & Gupta, A. K. (2026). Multi-Hazard Susceptibility Mapping for India using CNN U-Net and Adaboost Deep Learning, 4th International Conference on Resilience, Multi-Risk Assessment and Monitoring (ICONREM), June 16-19, 2026, Florence, Italy (*accepted*)
- C5. Thakur, D. A. and **Mohanty, M. P.*** (2026) Unraveling Seasonal Disparities in Flood Risks: Contrasting Monsoon and Post-Monsoon Impacts over Coastal Multi-Hazard Catchments, 10th International Conference on Flood Management 2026, May 20-26, 2026, Ontario, Canada.
- C6. Deopa, R., Debasish, M., Singh, H., Dayal, D., & **Mohanty, M. P.*** (2026). Integrated modelling of the flood-hydro-ecological-health nexus under future climate extremes in urban megacities. JpGU General Assembly 2026, Chiba, Japan, 24-29 May 2026.
- C7. Tripathi, V. and **Mohanty, M. P.*** (2026) Machine Learning-Driven Classification and Prediction of Flood Risks: Integrating Hydro-Climatic and Geomorphic Attributes for Effective Flood Management, 10th International Conference on Flood Management 2026, May 20-26, 2026, Ontario, Canada
- C8. Saha, S., Mukesh, S., Pandey, A., Rao, B. S., & **Mohanty, M. P.*** (2026) Assessing the Sensitivity of Multi-Source Topographic Data for Glacial Lake Outburst Flood (GLOFs) Risk Assessment, 4th Roorkee Water Conclave, 23-25, 2026, Roorkee, Uttarakhand, India (*received best paper award*)
- C9. Rachit, **Mohanty, M. P.***, Pandey, A. & Gupta, A. K. (2026) Multi-Hazard Susceptibility Mapping for India using Deep Learning, 4th Roorkee Water Conclave, Feb 23-25, 2026, Roorkee, Uttarakhand, India (*accepted*)
- C10. Tripathi, V., Shaw, Akshat, & **Mohanty, M.P.*** (2025). Nationwide Daily Baseflow Estimation in India (1980–2020) Using Temporal Fusion Transformers and Process-Informed Clustering, AGU Annual Meeting, December 15-19, 2025, New Orleans, USA.
- C11. Tripathi, V., Karmakar, S., Aghakouchak, A., **Mohanty, M.P.*** & Stein, L. (2025). Floods at the Crossroads of Climate and Human Impact: A Process-Based, Explainable AI Framework for Indian Catchments, AGU Annual Meeting, December 15-19, 2025, New Orleans, USA.
- C12. Tripathi, V., and **Mohanty, M.P.*** (2025). Interpretable AI for Flood Process Attribution and Prediction in Indian Catchments under Climate and Anthropogenic Influence, XIIth Scientific Assembly of the International Association of Hydrological Sciences (IAHS) 2025, October 05 – 10, 2025. Roorkee, India.
- C13. Saha, S., Pandey, A., Rao, B. S., & **Mohanty, M. P.*** (2025). Integrated Web-Based Framework for Risk Simulation and Visualization of Potentially Dangerous Glacial Lakes in Uttarakhand, AGU Annual Meeting, December 15-19, 2025, New Orleans, USA.
- C14. Rachit, Tripathi, V., **Mohanty, M. P.***, Pandey, A. & Gupta, A. K. (2025). Integrated Multi-Hazard Susceptibility Mapping across India using U-net Deep Learning Architecture, AGU Annual Meeting, December 15-19, 2025, New Orleans, USA, <https://agu25.ipostersessions.com/default.aspx?s=94-97-DE-BE-BB-2F-02-A2-66-5C-E6-A9-30-18-78-22>.
- C15. Rachit, **Mohanty, M. P.***, Pandey, A. & Gupta, A. K. (2025). Advancing Multi-Hazard Susceptibility Mapping Using U-Net Convolutional Neural Networks for India's Diverse

and Complex Terrain, Indian Symposium on Machine Learning (IndoML) 2025, Dec 18-21, 2025, Hyderabad, India. *(received a student travel grant to present)*

- C16. Tripathi, V., & **Mohanty, M.P.*** (2025). Interpretable AI for Flood Process Attribution and Prediction in Indian Catchments under Climate and Anthropogenic Influence, 5th International Vietnam Conference on Earth and Environmental Sciences 2025 (iVCEES-2025). *(received a student travel grant to present)*
- C17. Saha, S., Pandey, A., Rao, B. S., & **Mohanty, M. P.*** (2025). Development of a GLOF forecasting system through a novel concept of pre-simulated library over the Hindu Kush Himalaya region, EGU General Assembly 2025, Vienna, Austria, 27 Apr–2 May 2025, EGU25-15646, <https://doi.org/10.5194/egusphere-egu25-15646>
- C18. Tripathi, V., & **Mohanty, M.P.*** (2026) Machine Learning-Driven Classification and Prediction of Flood Risks: Integrating Hydro-Climatic and Geomorphic Attributes for Effective Flood Management, 10th International Conference on Flood Management 2026, May 20-26, 2026, Ontario, Canada *(accepted)*
- C19. Saha, S., Pandey, A., Rao, B. S., & **Mohanty, M. P.*** (2025). Development of a GLOF forecasting system through a novel concept of pre-simulated library over the Hindu Kush Himalaya region, EGU General Assembly 2025, Vienna, Austria, 27 Apr–2 May 2025, EGU25-15646, <https://doi.org/10.5194/egusphere-egu25-15646>
- C20. Tripathi, V., Deopa, R., & **Mohanty, M. P.*** (2025) Unraveling Hydrometeorological Drivers of Floods: A Data-Driven Analysis Across India's River Catchments, EGU General Assembly 2025, Vienna, Austria, 27 Apr–2 May 2025, EGU25-15096, <https://doi.org/10.5194/egusphere-egu25-15096>
- C21. Singh, H. & **Mohanty, M. P.*** (2025) Suitability of Reanalysis products in capturing Flood Inundation and Hazards over India: Deriving insights through Statistical tests and Numerical Flood Modeling , EGU General Assembly 2025, Vienna, Austria, 27 Apr–2 May 2025, EGU25-755, <https://doi.org/10.5194/egusphere-egu25-755>
- C22. Deopa, R., Mishra, D., Kumar Shahi, N., Tripathi, V., & **Mohanty, M. P.*** (2025) Quantifying human health risks from contaminated urban floodwaters using a multi-model framework under climate and socio-economic scenarios, EGU General Assembly 2025, Vienna, Austria, 27 Apr–2 May 2025, EGU25-15444, <https://doi.org/10.5194/egusphere-egu25-15444>
- C23. Thakur, D. A. and **Mohanty, M. P.*** (2025) How Divergent Are Flood Risks During Monsoon and Post-Monsoon Seasons? Revealing Contrasting Impacts over Coastal Multi-Hazard Catchments, EGU General Assembly 2025, Vienna, Austria, 27 Apr–2 May 2025, EGU25-418, <https://doi.org/10.5194/egusphere-egu25-418>.
- C24. **Mohanty, M.P.*** and Thakur, D. A. (2025) MANAGE: A Consolidated End-To-End Multi-Hazard Web-Based Flood Risk Information System for Coastal Regions, EGU General Assembly 2025, Vienna, Austria, 27 Apr–2 May 2025, EGU25-468, <https://doi.org/10.5194/egusphere-egu25-468>.
- C25. Rachit, Tripathi, V., **Mohanty, M. P.***, Pandey, A., Gupta, A. K. (2025). Multi-Hazard Susceptibility Mapping over India through Deep Learning Technique, XIIth Scientific Assembly of the International Association of Hydrological Sciences (IAHS) 2025, October 05 – 10, 2025. Roorkee, India *(accepted)*
- C26. Singh, H., Tyagi, M. & **Mohanty, M. P.*** (2025) A Novel High-Resolution Flood Hazard Model for India: Integrating Reanalysis Data with Hydrodynamic Simulations, XIIth IAHS Scientific Assembly 2025, Oct 05 - Oct 10, IIT Roorkee, India

- C27. **Mohanty, M. P.***, Saha, S., & Ojha, C. S. P. (2025). Development of a GLOF inundation library to strengthen near real-time flood risk management: An application over a sensitive glacial lake in India, International Perspectives on Water and Environment (IPWE), Bhopal, 8-10 January 2025.
- C28. Deopa, R., & **Mohanty, M. P.*** (2024). Quantifying the Impact of Urbanization and Rising Rainfall Extremes on Flood Risks in Delhi through Spatio-Temporal Analysis and Hydrodynamic Modeling. Young Leaders Conclave (YLC) 2024, Jun. 04-05, 2024, Delhi, India. *(received best paper award)*
- C29. **Mohanty, M. P.***, Tyagi, M., Singh, H., and Thakur, D. (2024). A novel Bivariate Risk Classifier to capture Compound Risks from Flood and Droughts over Multi-hazard coastal catchments, Multi-hazard DRR Academy for young researchers and practitioners, Barcelona, Spain, 23-26 October 2024 *(received a travel grant to present and convene a session)*
- C30. **Mohanty, M. P.***, and Deopa, R. (2024). An integrated framework to bridge and map flood risks and human health risks from contaminated floodwaters during urban flooding episodes, 9th International Symposium on Integrated Water Resources Management (IWRM), 4-7 November 2024, Florianopolis, Brazil [*received the SYSTA travel grant from IAHS*].
- C31. **Mohanty, M. P.***, and Deopa, R. (2024). Assessment of the microbial risk in contaminated flood waters: Implications of climate and socio-economic dynamics, American Chemical Society Fall 2024, 21 August 2024.
- C32. Tripathi, V., & **Mohanty, M. P.*** (2024). A Hybrid Framework of integrating Geomorphic Flood Descriptors and Machine Learning Models to account for Flood Risk over an ungauged basin, HIWeather Final Conference 2024, 09 September 2024 to 13 September 2024. Munich, Germany [*received a student travel grant from the World Meteorological Organization, WMO*]
- C33. Singh, H., & **Mohanty, M. P.*** (2024). Suitability of Reanalysis products in capturing Flood Inundation and Hazards over India: Deriving insights through Statistical tests and Numerical Flood Modeling, 6th WCRP International Conference on Reanalysis (ICR6), 28th October 2024 to 1st November 2024. Tokyo, Japan
- C34. Thakur, D.A., Singh, H., & **Mohanty, M. P.*** (2023) Fidelity of RCMs in Quantification of Climate Change Impacts on Flood Risk- An exhaustive approach over severely flood-prone coastal catchment, International Conference on Regional Climate ICRC-CORDEX 2023, 25 September 2023 to 29 September 2023. Trieste, Italy *(received a student travel grant for presentation in Italy)*
- C35. Tripathi, V., **Mohanty, M. P.*** (2023). Leveraging CORDEX simulations with Geomorphic Flood Descriptors: A Machine Learning based approach for flood risk estimation over the Ganga sub-basin, International Conference on Regional Climate ICRC-CORDEX 2023, 25 September 2023 to 29 September 2023. Trieste, Italy
- C36. Deopa, R., **Mohanty, M. P.*** (2023). Integrating regional climate models in flood modeling to quantify human health risks from contaminated urban floodwaters, International Conference on Regional Climate ICRC-CORDEX 2023, 25 September 2023 to 29 September 2023. Trieste, Italy *(received a student travel grant for presentation)*
- C37. Singh, H., & **Mohanty, M. P.*** (2023) Checking the performance of Regional Climate models in determining Flooding at Basin Scale using Global Hydrodynamic model, International Conference on Regional Climate ICRC CORDEX-2023, 25 September 25 2023 to 29 September 2023. Trieste, Italy *(received a student travel grant for presentation)*
- C38. Thakur, D.A., Singh, H., & **Mohanty, M. P.*** (2023) An Amalgamated Approach to quantify Climate Change Impacts on Flood Risk - Utilisation of RCMs for Efficient Flood Management at Regional Scale, WCRP Open Science Conference 2023, October 23-27, 2023, Rwanda.

- C39. Singh, H., & **Mohanty, M. P.*** (2023). Is ERA5 able to reproduce flood inundation at the regional scale? A systematic analysis using LISFLOOD-FP, 12th World Congress on Water Resources and Environment (EWRA 2023), June-July, 27- 01, 2023, Thessaloniki, Greece.
- C40. Singh, H., Tripathi, V., & **Mohanty, M. P.*** (2023). Can Regional Climate Models represent flood hazards at regional scales?: Attributing performance through global flood models over a data-scarce flood-prone region in India, WCRP Open Science Conference 2023, October 23-27, 2023, Rwanda.
- C41. Tripathi, V., Singh, H., **Mohanty, M.P.*** (2023). Fidelity of Machine Learning models in capturing flood inundation through geomorphic descriptors over Ganga sub-basin, India. The 12th World Congress European Water Resources Association (EWRA 2023), June-July 27- 01, 2023, Thessaloniki, Greece.
- C42. Tripathi, V., **Mohanty, M.P.*** (2023). Employing CORDEX-RCMs for augmenting the performance of Geomorphic Flood Descriptors over large flood-prone and ungauged watersheds. WCRP Open Science Conference 2023, October 23-27, 2023, Rwanda.
- C43. Deopa, R., Tripathi, V., **Mohanty, M.P.*** (2023). Capturing the teleconnections of climate change and socio-economic structures on flood and human health risks from contaminated urban flood. WCRP Open Science Conference 2023, October 23-27, 2023, Rwanda.
- C44. Shirsath, V., Thakur, D. A., & **Mohanty, M. P.*** (2023). Identifying Nuisance flood risks by aggregating hazard and vulnerability dimensions over a severely flood-prone urbanized catchment in India, WCRP Open Science Conference 2023, October 23-27, 2023, Rwanda
- C45. Namgyal, T., **Mohanty, M. P.***, and Thakur D. A., (2023). How fitting are open-source flood models in capturing flood risks over mountainous regions: A prudent analysis over Chamkar Chu Basin, Bhutan, using HEC-RAS v6.3. EGU General Assembly 2023, Vienna, Austria, 24–28 Apr 2023.
- C46. **Mohanty, M.P.***, & Simonovic, S. P. (2023) Quantifying the changes in floodplain regimes over Canada under changing climate conditions- Development of Flood Map Viewer to enhance disaster preparedness, 9th International Conference on Flood Management (ICFM9), February 18-22, 2023, Tsukuba, Japan (*received a travel grant for presentation*)
- C47. Thakur, D.A., & **Mohanty, M. P.*** (2022) Modeling Multi-Hazard Coastal Flood Risk with Satellite Precipitation Product: A Synergistic Effort to Quantify Impacts at the Inundation Scale, AGU Fall Meeting, December 12-16, 2022, Chicago, USA (*received a student travel grant for presentation*)
- C48. Thakur, D.A., & **Mohanty, M. P.*** (2022) Efficacy of CHIRPS V2.0 in Characterizing Flood Hazards in a Resource Constrained Coastal Catchment, International Conference on Climate and Weather-Related Extremes (ICCWE2022), September 19-20, 2022, Roorkee, India (*received the Best Paper Award*)
- C49. **Mohanty, M.P.***, & Simonovic, S. P. (2022) Flood Map Viewer- Development of a web-based high-resolution flood hazard information system for flood risk management in Canada, International Conference on River Corridor Research and Management, May 30-June 1, 2022, Guwahati, India.
- C50. **Mohanty, M.P.***, & Simonovic, P. S. (2021) A comprehensive approach for flood-plain mapping through identification of hazard using publicly available data sets over Canada, ICFM 8, Aug. 17-19, 2021, Iowa, USA. (*received travel grant for presentation*)
- C51. **Mohanty, M.P.**, & Karmakar, S.* (2021) Tide Equivalent Rainfall: A new metric to improve the resilience of global coastal regions to pluvial and tidal impacts, ICFM 8, Aug. 17-19, 2021, Iowa, USA. (*received a travel grant for presentation*)

- C52. **Mohanty, M.P.**, Karmakar, S.*, & Ghosh, S. (2019) WEBFRIS: a web-based flood information system for enhanced resilience through risk awareness, EWRA 2019, Jun. 23-25, 2019, Madrid, Spain. *(received a travel grant for presentation)*
- C53. **Mohanty, M.P.**, Vittal, H., Karmakar, S.*, & Ghosh, S. (2017). An exhaustive approach for identification of flood risk hotspots in data-poor regions enforcing combined geomorphic and socio-economic indicators, Paper Number: NH21B-0174, AGU Fall Meeting, Dec. 11-15, 2017, New Orleans, USA.
- C54. Singh, J., Paimazumder, D., **Mohanty, M.P.**, Karmakar, S.*, & Ghosh, S. (2017). Urbanization induces non-stationarity in Extreme Rainfall Characteristics over Contiguous United States, Paper Number: GC43F-1125, AGU Fall Meeting 2017, Dec. 11-15, 2017, New Orleans, USA.
- C55. Sharma, T., Vittal, H., **Mohanty, M.P.**, Ghosh, S., & Karmakar, S.* (2017). Assessing the Impacts of Decadal Socio-Agro-Hydro Climatic Variations on Agricultural Vulnerability over India. Paper Number: GC34B-08, AGU Fall Meeting 2017, Dec. 11-15, 2017, New Orleans, USA.
- C56. **Mohanty, M.P.**, Nithya, S., Indu, J., Karmakar, S.* & Ghosh, S. (2017) Impacts of different sources of topographic data and uncertainty assessment on 1-D 2-D coupled flood inundation modelling, United Nations International Conference on Space-based Technologies for Disaster Risk Reduction - "Building Resilience through Integrated Applications", Oct. 23-25, 2017, Beijing, China.
- C57. **Mohanty, M.P.**, Gusain, A., Sharma, T., Ranjan, P., Karmakar, S.*, & Ghosh, S. (2016). A Comparative Flood Hazard Assessment with Climate Projections for India, ICRC: Cordex 2016, May 17-20, 2016, Stockholm, Sweden. *(received a travel grant for presentation)*
- C58. **Mohanty, M.P.**, Gusain, A., Sherly, M.A., Karmakar, S.*, & Ghosh, S. (2016). Estimation of Design rainfall for flood risk mapping in a data-scarce situation, 9th International Geographical Union Conference, Mar. 18-20, 2016, New Delhi, India
- C59. Gusain, A., **Mohanty, M.P.**, Vittal, H., Karmakar, S.*, Ghosh, S., & Rao, G.S (2015) Assessment of future change in streamflow through the identification of its association with precipitation in Mahanadi Basin, India under the data-scarce scenario, Paper Number: H43I-1676, AGU Fall Meeting 2015, Dec. 14-18, 2015, San Fransisco, USA.
- C60. **Mohanty, M.P.**, Brahmacharimayum, B., & Ghosh, P.K.* (2014). Simultaneous bio-reduction of sulfate and nitrate in batch shake flasks Poster presentation at National Conference on Sustainable Development of Environmental Systems, Jun. 20-21, 2014, Guwahati, India.
- C61. **Mohanty, M.P.**, & Ghosh, P.K.* (2014). A review on various Bio-kinetic models used in biological sulfate reduction, International Conference on Harnessing Natural Resources for Sustainable Development – A global trend (IHNRSD-14), Jan. 29-31, 2014, Guwahati, India.
- C62. Brahmacharimayum, B., **Mohanty, M.P.**, & Ghosh, P.K.* (2014). A review on the intermediates, pathways, and enzymes utilized in dissimilatory sulfate reduction, International Conference on Harnessing Natural Resources for Sustainable Development – A global trend (IHNRSD-14), Jan. 29-31, 2014, Guwahati, India.
- C63. Laskar, M., **Mohanty, M. P.**, Brahmacharimayum, B., & Ghosh, P.K.* (2013). Kinetics of sulfate bio-reduction and sulfide oxidation by mixed microbial consortia, International Congress of Environmental Research (ICER-2013), Dec. 19-21, 2013, Aurangabad, India.

BOOK/EDITED VOLUME

1. Yadav, B.*, **Mohanty, M.P.**, Pandey, A., Singh, V.P., and Singh, R.D. (2022). Sustainability of Water Resources - Impacts and Management, Springer Nature <https://doi.org/10.1007/978-3-031-13467-8>
2. Khatua, K.K.*, **Mohanty, M.P.**, & Patel, M. (2026). Coastal and Ocean Engineering Vol 3, Springer Nature [in progress]

BOOK CHAPTERS

1. Rachit, **Mohanty, M. P.***, Pandey, A., & Gupta, A. K. (2025). Managing Disasters Using Geospatial Technology and AI: A Multi-Hazard Perspective. In *Handbook of Artificial Intelligence-driven Digital Image Analysis for Intelligent Remote Sensing*. CRC Press, Taylor & Francis Group (accepted, in press)
2. Thakur, D. A., & **Mohanty, M. P.*** (2025). Utilizing Multi-criteria Decision-Making Techniques for Computing Composite Vulnerability over a Severely Flood-Prone Multi-hazard Catchment in *Progress in Multicriteria Decision Making Models*, Springer, ISBN: 978-3-031-89245-5.
3. Singh, H., Tripathy, P., Thakur, D.A., & **Mohanty, M. P.*** (2025) Socioeconomic-Cum-Physical Vulnerabilities to Riverine Floods over Large Watersheds: A Comprehensive Framework by Leveraging Shannon Entropy and TOPSIS in *Progress in Multicriteria Decision Making Models*, Springer, ISBN: 978-3-031-89245-5.
4. Tyagi, M., & **Mohanty, M. P.*** (2025) GIS-Based Multi-Hazard Risk Analysis and Disaster Risk Reduction Planning Using Coupled Multi-Hazard Assessment Method in *Progress in Multicriteria Decision Making Models*, Springer, ISBN: 978-3-031-89245-5.
5. Thakur, D. A., Suryawanshi, V., Ramesh, H., & **Mohanty, M. P.*** (2025). Evaluating the reliability of open-source hydrodynamic models in flood inundation mapping: an exhaustive approach over a sensitive coastal catchment. In *Hydrosystem Restoration Handbook* (pp. 93-108). Elsevier. <https://doi.org/10.1016/B978-0-443-29802-8.00007-8>
6. Barman, D., Verma, R. K., Rongali, R., **Mohanty, M. P.**, and Pandey, A.* (2025). Impacts of Land Use/Land Cover and Climate Change on Soil Erosion - A Watershed-Based Approach, Hydro-Environmental Impact of Climate and Land Use Change on Watersheds for Sustainable Development Cambridge Scholars Publishing, 304-328.
7. Karmakar, S.*, Sherly, M. A., & **Mohanty, M.P.** (2022) Urban Flood Risk Mapping: A state-of-the-art review on quantification, current practices, and future challenges, *Advances in Urban Design and Engineering*, Springer, 125-156 https://doi.org/10.1007/978-981-19-0412-7_5
8. **Mohanty, M.P.**, and Karmakar, S.* (2022) Flood risk estimation and mapping: Present status and future challenges *Mathematical Advances towards Sustainable Environmental Systems-Earth Systems Protection*. Springer <https://doi.org/10.1007/978-3-030-98584-4>

MAJOR TECHNICAL REPORTS

1. **Mohanty, M.P.***, Pandey, A., & K, Munish. (2025). Interim report on the State Specific Action Plan for the Water Sector in Odisha. A report was submitted to the Department of Water Resources, Govt. of Odisha, in December 2025.
2. **Mohanty, M.P.***, Pandey, A., & K, Munish. (2025). Status report on the State Specific Action Plan for the Water Sector in Odisha. A report was submitted to the Department of Water Resources, Govt. of Odisha, in May 2025.
3. **Mohanty, M.P.*** (2025). An integrated approach to quantify the impacts of climate change and socioeconomic dynamics on human health risk associated with contaminated urban floodwaters. A report was submitted to the Anusandhan National Research Foundation (ANRF) in March 2025.

4. **Pandey, A.,** & Mohanty, M.P. (2024). Identification of Critical Soil Erosion prone areas and Preparation of the CAT Plan. A report was submitted to the National Mission for Clean Ganga, Ministry of Jal Shakti, Govt. of India, in March 2024.
5. **Mohanty M.P.*,** Simonovic, S.P. (2021). A generic framework to quantify changes in floodplain regimes by incorporating climate change impacts over large regions. Water Resources Research Report, Report No. 109, ISBN: 978-0-7714-3157-9 (print); 978-0-7714-3158-6 (online).
6. **Mohanty M.P.*,** Simonovic, S.P. (2020) A comprehensive framework for regional floodplain mapping, Water Resources Research Report, Report No. 109, ISBN: 978-0-7714-3147-0 (print); 978-0-7714-3148-7 (online).
7. Bogdana S., **Mohanty M.P.*,** & Simonovic, S.P. (2020). Regional analysis of population exposure to flooding in Canada, Water Resources Research Report, Report No. 110, ISBN: 978-0-7714-3151-7 (print); 978-0-7714-3152-4 (online).
8. **Mohanty M.P.*,** Ghosh, S., & Karmakar, S. (2019). A comprehensive mapping of flood risk in a changing climate: An application to Jagatsinghpur District, Odisha: A report submitted to the National Remote Sensing Centre (NRSC), Hyderabad in Jul. 2018.

LIST OF DESIGNS, PRODUCTS, PROCESSES, AND PROJECT IMPLEMENTATION

- HyEco- A hybrid hydrodynamic-cum-ecological modeling framework to capture the dynamics of the spread of contaminated waters during flood wave propagation
- MANAGE (**M**ulti-h**A**zard Flood information**N** system for co**A**stal re**G**ion): The comprehensive web-repository of flood hazards, physical and socio-economic vulnerabilities, and bivariate flood risks at the finest administrative scale derived through a cascade of high-resolution flood modeling, Multi-Criteria Decision-Making tools, and statistical techniques (accessible at <https://manage.wr.iitr.ac.in/>).
- Flood map viewer for Canada- A first-of-its-kind web-based information system of Canadian floodplain maps for historical and future periods, incorporating climate change impacts at 1 km resolution, was developed. The system is now used by the federal govt., policymakers, and disaster management experts (accessible at <https://www.floodmapviewer.com/>).
- Bivariate flood risk classifier- To provide precise identification of marginal and compound contributions of hazard and vulnerability to overall flood risk at the finest administrative scale of village level
- Tide Rainfall Flood Quotient (TRFQ)- The new metric was demonstrated over Mithi catchment, Jagatsinghpur, and Chennai to determine the propensity of coastal flooding due to rainfall and storm tide.

LIST OF TECHNOLOGY TRANSFERS

1. **Mohanty, M.P.,** & Deopa, R. (2025). HyEco- A hybrid hydrodynamic-cum-ecological modeling framework to capture the dynamics of the spread of contaminated waters during flood wave propagation. Detailed report was submitted in March 2025.
2. **Mohanty, M.P.,** Ghosh, S., & Karmakar, S. (2019). A comprehensive mapping of flood risk in a changing climate: An application to Jagatsinghpur District, Odisha. Technology was transferred, and the report was submitted to the National Remote Sensing Centre (NRSC), Hyderabad, in Jul. 2018.
3. Ghosh, S., Karmakar, S., Saha, A., **Mohanty, M.P.** et al. (2019). Development of India's first integrated expert urban flood forecasting system for Chennai. Technology was transferred, and the report was submitted to the Scientific Advisor of the Prime Minister's Office in 2019.

MEDIA COVERAGES (selected and relevant)

1. "Extreme rain triggered cascading disasters in western Himalayas: IIT-R study" in the Times of India, Oct. 1, 2025 (Complete news- <https://timesofindia.indiatimes.com/city/dehradun/extreme-rain-triggered-cascading-disasters-in-western-himalayas-iit-r-study/articleshow/124263739.cms>)
2. "IIT Roorkee develops tool to track flood-borne diseases, tested in Delhi floods" in India Today, Aug. 20, 2025 (Complete news- <https://www.indiatoday.in/education-today/news/story/iit-roorkee-develops-tool-to-track-flood-borne-diseases-tested-in-delhi-floods-2774153-2025-08-20>)
3. "HyEco: A hybrid hydrodynamic-cum-ecological modelling framework to demystify flood and human-health risks over flood-prone metropolis" in the Prevention Web, UN Office for Disaster Risk Reduction (UNDRR), Aug. 12, 2025 (Complete news- <https://www.preventionweb.net/publication/documents-and-publications/hyeco-hybrid-hydrodynamic-cum-ecological-modelling-framework>)
4. "Do disparities exist in flood risks during monsoon and post-monsoon seasons? Comprehending diametric behaviors over coastal multi-hazard catchments" in the Prevention Web, UN Office for Disaster Risk Reduction (UNDRR), Jan. 13, 2025 (Complete news- <https://www.preventionweb.net/publication/do-disparities-exist-flood-risks-during-monsoon-and-post-monsoon-seasons-comprehending>).
5. "IIT Roorkee's study empowers communities to tackle flood risks amid cyclones" in India Today, Dec. 1, 2024 (Complete news- <https://www.indiatoday.in/education-today/news/story/iit-roorkees-study-empowers-communities-to-tackle-flood-risks-amid-cyclone-fengal-2643092-2024-12-01>)
6. "Delhi's 2023 floods linked to rapid urbanisation, climate change: IIT Roorkee" in the India Today, Sept. 17, 2024 (Complete news- <https://www.indiatoday.in/education-today/news/story/iit-roorkee-study-delhi-floods-linked-urbanisation-climate-change-2600541-2024-09-16>)
7. "Why India Needs Better Flood Preparedness" in the IndiaSpend, Aug. 29, 2023 (Complete news- <https://www.indiaspend.com/earthcheckindia/why-india-needs-better-flood-preparedness-874673#:~:text=India's%20geographical%20landscape%2C%20changing%20weather,India%20of%20the%20Indian%20Space>)
8. "Researcher develops Canadian floodplain map to show impact of climate change flooding" in the Toronto Star, Nov. 2, 2021. (Complete news- <https://www.thestar.com/news/canada/2021/11/03/researcher-develops-canadian-floodplain-map-to-show-impact-of-climate-change-flooding.html?fbclid=IwAR18p2NorjGCjBIAYNk7GM2TM7iYfw1MhKxRGM5ZyuECTzT7MQH3LLIZcL0>).
9. "Flood-impact map-a Canadian first" in the Western News, Nov. 3, 2021. (Complete news- https://news.westernu.ca/2021/11/flood-impact-map-a-canadian-first/?fbclid=IwAR124w3umBwELfjpyzgjqoRFORGwDV0iQCjbIGHFe_M_rAb4dec7t09bHY).
10. "New flood risk prediction tool to help "map" out disaster mitigation plans" in the Prevention Web, UN Office for Disaster Risk Reduction (UNDRR), Aug. 31, 2021. (Complete news- <https://www.preventionweb.net/news/new-flood-risk-prediction-tool-help-map-out-disaster-mitigation-plans>) .
11. "A new measure that can predict whether a city is prone to rain- or tide-based flooding" in The Hindu, Jan. 02, 2021. (Complete news- <https://www.thehindu.com/sci-tech/science/a-new-measure-that-can-predict-whether-a-city-is-prone-to-rain-or-tide-based-flooding/article33482930.ece?homepage=true>).
12. "Finding flood risks at the village-level to facilitate decision-making" in The Planet Outlook, Oct. 11, 2020. (Complete news-

<https://planet.outlookindia.com/water/finding-flood-risks-at-the-village-level-to-facilitate-decision-making/594>)

13. "New Recommendations for a Proactive Flood Policy in India" in the EoS, Science News by American Geophysical Union, Sept. 17, 2020. (Complete News- <https://eos.org/articles/new-recommendations-for-a-proactive-flood-policy-in-india>)
14. "How to save Jagatsinghpur from floods" in the Odisha Post, Sept. 14, 2020. (Complete News- <http://odishapostpaper.com/edition/967/orissapost/page/3>)
15. "Flood risk on the rise: An urgent need of a socio-technocratic approach for flood management" in the Political and Business Daily, Feb. 16, 2020.
16. "New study offers promising way of looking at flood risk", in the Odisha Post, Jan. 2, 2020. (Complete news- https://www.orissapost.com/new-study-offers-promising-way-of-looking-at-flood-risk/?fbclid=IwAR2MNd2s3hiyTmGvJg-gYjkzV2Pvxs2Wg_1SrKL1krfL6-wA3ZGTYXom390)
17. "Indian scientists prepare first urban flood forecasting" in the Deccan Herald, Oct. 13, 2019. (Complete news- <https://www.deccanherald.com/national/indian-scientists-prepare-first-urban-flood-forecasting-768178.html>)

SPONSORED RESEARCH PROJECTS

S.No.	Title of Project	Funding Agency	Financial Outlay	Year of start & total period	Details of other investigators	Status
As PI and Co-PI						
1	Rethinking Disaster Management- A multi-hazard approach of floodplain mapping over a coastal region in India	Faculty Initiation Grant, SRIC, IIT Roorkee	20 Lakhs	8 Oct. 2021 to 10 Oct. 2023; 2 years	PI: Prof. Mohit P. Mohanty	Completed
2	FLOOD-TWIN: A Comprehensive Digital Twin Framework Integrating Forecasting and Hydrodynamic Modeling for Urban Flood Management	National Institute of Disaster Management (NIDM), Govt. of India	7.98 Lakhs	30 April 2026 to 30 April 2028	PI: Prof. Mohit P. Mohanty Co-PI: Prof. Ashish Pandey Contact from NIDM: Dr. Gagandeep Singh	In progress
3	An integrated approach to quantify the impacts of climate change and	ANRF (Previously SERB), Govt. of India	27.71 Lakhs	01 Nov. 2022 to 31 Jan. 2025;	PI: Prof. Mohit P. Mohanty	Completed

S.No.	Title of Project	Funding Agency	Financial Outlay	Year of start & total period	Details of other investigators	Status
As PI and Co-PI						
	socioeconomic dynamics on human health risk associated with contaminated urban floodwaters			2 years 3 months		
4	Development of a web-based GLOF information system: An integrated approach for quantifying flood hazards over Safed Lake, Uttarakhand	Indian Space Research Organization (ISRO)	25.44 Lakhs	27 Feb. 2024 to 28 Feb. 2027. 3 years	PI: Prof. Mohit P. Mohanty Co-PI: Prof. Ashish Pandey	In progress
5	Multi-hazards riskscapes and resilience pathways of coastal fishing communities (R-Path)	National Institute of Disaster Management (NIDM), Govt. of India	8.00 Lakhs	30 April 2026 to 30 April 2028	PI: Prof. Roopam Shukla Co-PI: Prof. Mohit P. Mohanty Contact from NIDM: Dr. Navneet Kumar	In progress
6	Comprehensive Climate Adaptation Strategy and a Detailed Implementation Roadmap on Disaster Management and Infrastructure Resilience for India's National	Ministry of Environment, Forest and Climate Change (MoEFCC), Govt. of India, and United Nations Development	47.74 Lakhs	03 Apr. 2025 to 03 Apr. 2027; 2 years	PI: Prof. Anil Gupta Co-PI: Prof. Mohit P. Mohanty, Prof. Ashish Pandey	In progress

S.No.	Title of Project	Funding Agency	Financial Outlay	Year of start & total period	Details of other investigators	Status
As PI and Co-PI						
	Adaptation Plan (Adapt-DMIR)	Programme (UNDP)				
7	Assessment of Environmental Flows in the Ghagra and Gomti River Basins	National Mission for Clean Ganga, Department of Water Resources, River Development & Ganga Rejuvenation, Govt. of India	842.73 Lakhs	06 Jun. 2025 to 06 Jun. 2028; 3 years	PI: Prof. Ashish Pandey Co-PIs: Prof. Mohit P. Mohanty, Prof. Basant Yadav, Prof. Kritika Kothari, Prof. Ellora Padhi, Prof. S.K. Mishra, Prof. Anshul Yadav	In progress
8	Quantifying infant health risks from urban flooding in India: An integrated hydrodynamic-epidemiological modeling approach	Stanford University, USA	82.25 Lakhs	1 June 2026 to 31 May 2028; 2 years	PI: Prof. Subhankar Karmakar, IITB Co-PI: Prof. Subimal Ghosh, IITB, Prof. Mohit P. Mohanty	Accepted, In progress
9	Identification of critical soil erosion-prone areas and preparation of the CAT plan	National Mission for Clean Ganga, Department of Water Resources, River Development & Ganga Rejuvenation, Govt. of India	40.40 Lakhs	January 2022 3 years	Pi: Prof. Ashish Pandey Co-PI: Prof. Mohit P. Mohanty	Completed

S.No.	Title of Project	Funding Agency	Financial Outlay	Year of start & total period	Details of other investigators	Status
As PI and Co-PI						
10	PARTNER: SDG – Partnership for an Equitable Future: Resilient Food and Water Systems under Global Change project	DAAD: German Academic Exchange Service	392 Lakhs (390, 250.79 €)	01 Feb. 2025 to 31 Dec. 2028 4 years	PI (Germany): Prof. Nicola Fohrer, Univ. of Kiel, Germany PI (India): Prof. Ashish Pandey Co-PIs (India): Prof. Mohit P. Mohanty, Prof. Basant Yadav, Prof. Kritika Kothari Co-PIs (Sri Lanka): Prof. T.N. Wickramaarachchi, Prof. Lishani Nisansala Wijewardene, Prof. Asanka Jayasinghe	In progress
11	Potential for scaling up indigenous rainwater harvesting techniques in semi-arid regions of India	Natural Environment Research Council	34.5 Lakhs (28, 980 £)	Jan. 2022 to July 2022 6 months	PI (India): Prof. Basant Yadav PI PI (UK): Prof. Alison Parker Co-PIs (India): Prof. Mohit P. Mohanty Co-PIs (UK):	Completed

S.No.	Title of Project	Funding Agency	Financial Outlay	Year of start & total period	Details of other investigators	Status
As PI and Co-PI						
					Prof. Jerry Knox, Prof. Abdou Khouakhi	
12	Research Collaborator for the Work Package-IV under the Integrated Centre for Adaptation to Climate Change, Disaster Risk Reduction, and Sustainability-IIT Roorkee (ICARS-IITR)	Department of Science and Technology (DST), Govt. of India	25.7 Lakhs allocated under the WP-IV (overall cost of the project is 672.05 Lakhs)	06 May 2024 to 06 May 2029 5 years	PI: Prof. Anil Gupta Co-PIs: Prof. K.K. Pant, Prof. Akshay Dwivedi, Prof. Ashish Pandey, Prof. Komal Tripathi	In progress
13	Developing Vulnerability Assessment for Climate Induced hazards/ Extreme Events for Uttarakhand (Jointly with STS Global)	Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ) GmbH	124 Lakhs	December 2025 to June 2026	PI: Prof. Anil K. Gupta Co-PI: Dr. Anshu Sharma, Founder SEEDS, Prof. K.K. Pant, Prof. Mohit P Mohanty, Prof. Roopam Shukla	Accepted, In progress

S.No.	Title of Project	Funding Agency	Financial Outlay	Year of start & total period	Details of other investigators	Status
As Research Collaborator						
14	Large-scale floodplain mapping using various publicly available datasets	Natural Sciences and Engineering Research Council (NSERC), Institute for Catastrophic Loss Reduction (ICLR), and Chaucer Syndicate	--	October 2019 to March 2021	PI: Prof. Slobodon P. Simonovic, Western University, Canada Post-Doctoral Fellow: Dr. Mohit P. Mohanty	Completed
15	Flood hazard mapping of data-scarce large flood-prone regions using geomorphic classifiers	Indian Space Research Organisation	--	June 2019 to August 2019	PI: Prof. Subhankar Karmakar, IITB Research Associate: Mohit P. Mohanty	Completed
16	A comprehensive mapping of flood risk in changing climate: An application to Jagatsinghpur district, Odisha (India)	Indian Space Research Organisation	--	January 2015 to May 2019	PI: Prof. Subhankar Karmakar, IITB Co-PI: Prof. Subimal Ghosh, IITB Ph.D research scholar: Mohit P. Mohanty	Completed
17	Design of an Expert System for Flood Forecasting and Management for the City of Chennai	Office of the Principal Scientific Adviser to the Prime Minister's Office, Govt. of India	--	January 2018 to May 2019	PI: Prof. Subimal Ghosh, IITB Co-PI: Prof. Subhankar Karmakar, IITB Associated research: Mohit P. Mohanty	

CONSULTANCY PROJECTS

S No.	Title of Project	Funding Agency	Financial Outlay	Year of start & total period	Details of other investigators	Status
1	State Specific Action Plan (SSAP) of Odisha	National Water Mission, Ministry of Water Resources, River Development and Ganga Rejuvenation, Govt. of India; and Department of Water Resources, Govt. of Odisha	82.60 Lakhs	February 2024 to December 2025 18 months	PI: Prof. Mohit P. Mohanty Co-PI: Prof. Ashish Pandey	In progress
2	Establishment of Design Basis Flood Level (DBFL) at Heavy Water Plant Manuguru	Department of Atomic Energy, Heavy Water Board, Govt of India	17.26 Lakhs	November 2025 to June 2026	PI: Prof. Mohit P. Mohanty	In progress
3	Rejuvenation of water bodies	Delhi Development Authority (DDA)		November 2025 to May 2026	PI: Prof. Basant Yadav Co-PI: Prof. Mohit P. Mohanty, Prof. Ashish Pandey	In progress
4	Training programme on water, built environment and emergency preparedness (Batch no. 1)	National Health Mission, Govt. of Odisha	23.48 Lakhs	16 April to 19 April 2024	PI: Prof. Mohit P. Mohanty Co-PIs: Prof. Basant Yadav Prof. Ashish Pandey	Completed
5	Training programme on water, built environment and emergency	National Health Mission, Govt. of Odisha	22.70 Lakhs	29 April to 2 May 2024	PI: Prof. Mohit P. Mohanty	Completed

S No.	Title of Project	Funding Agency	Financial Outlay	Year of start & total period	Details of other investigators	Status
	preparedness (Batch no. 2)				Co-PIs: Prof. Basant Yadav Prof. Ashish Pandey	
6	Training programme on water, built environment and emergency preparedness (Batch no. 3)	National Health Mission, Govt. of Odisha	21.92 Lakhs	6 May to 9 May 224	PI: Prof. Mohit P. Mohanty Co-PIs: Prof. Basant Yadav Prof. Ashish Pandey	Completed
7	Training programme on water, built environment and emergency preparedness (Batch no. 4)	National Health Mission, Govt. of Odisha	21.13 Lakhs	13 May to 16 May 2024	PI: Prof. Mohit P. Mohanty Co-PIs: Prof. Basant Yadav Prof. Ashish Pandey	Completed
8	International training programme on River engineering and management	Ministry of External Affairs	1.47 Lakhs	13 March to 15 March 2023	PI: Prof. Mohit P. Mohanty Co-PIs: Prof. Ashish Pandey	Completed
9	International training programme on Applications of Remote Sensing and GIS in Agriculture	Ministry of External Affairs	1.47 Lakhs	1 September to 3 September 2022	PI: Prof. Mohit P. Mohanty Co-PIs: Prof.	Completed

S No.	Title of Project	Funding Agency	Financial Outlay	Year of start & total period	Details of other investigators	Status
					Ashish Pandey	
10	Recent advances in satellite technology for monitoring hydro-climatological extremes	ISRO Disaster Management Support Programme	5.32 Lakhs	11 September to 15 September 2023	PI: Prof. Mohit P. Mohanty Co-PIs: Prof. Ashish Pandey	Completed

INVITED RESEARCH PRESENTATIONS

S No.	Year	Title of Lecture/ Lecture Series	Date, Place and Programme
	2025	Invited lecture on “Ascertaining Multi-Hazard Risks for Improved Resilience: Challenges and Lessons Learned in India”	19 November 2025 Recent Computational Advances in Water Resources Modelling and Management (RCA-WRMM-2025) at NIT Silchar
		Invited lecture on the “Rejuvenation of lakes.”	19 September 2025 Brainstorming session on Reimagining Lakes of Goa, Goa
1		HyEcO: An integrated framework to bridge Hydrodynamic and Ecological Risks during unprecedented climate extremes	19 August American Chemical Society Fall 2025 Digital Meeting - Global Virtual Symposia (Online)
2		Glacier Preservation in a Warming World: Science and Solutions	21 March 2025 Indian Water Resources Society, Meerut World Water Day 2025
3		Integrated Strategies for Flood Risk Reduction and Adaptive Water Governance	21 and 22 August 2025 Haridwar National Workshop on Adaptation Planning for India's Water Sector under Climate Change, sponsored by UNDP and Green Climate Fund, and organized by National Institute of Hydrology (NIH), Roorkee
4		Invited lecture on “Socio-Technocratic” approach towards effective Multi-hazard Risk Management: Current status and future challenges	28 July to 2 August 2025 Kiel University, Germany PARTNER - SDG Partnership for an equitable future workshop
5		Invited lecture on “HyEcO: An integrated framework to bridge hydrodynamic and ecological risks during unprecedented climate extremes”	5-14 August 2025 Kiel University, Germany SWAMP Summer School and Workshop Resilient and equitable water future
6		Invited lecture on “Urban Flood Risk Assessment in India: Challenges, Lessons, and the Way	7 April 2025 National Institute of Disaster Management, New Delhi, Ministry of Home Affairs,

S No.	Year	Title of Lecture/ Lecture Series	Date, Place and Programme
		Forward"	Government of India Training programme on Urban Floods: Risk Assessment, Mitigation, and Management
7		Invited talk on "Advancing Flood Risk Mapping Through Applications of Machine Learning Techniques and Geomorphic Classifiers"	17-28 February 2025 NIT Patna Faculty Development Programme on "Transforming Water Resources Management, Traffic Systems, and E-Mobility with Quantum Computing and AI"
8		Invited talk on "Flood Risk Modeling: From Hydrological to Human-health perspectives"	9-10 Dec. 2024 IIT Madras International Workshop on Flood Risk and Climate Change (IWFRCC) – 2024 (Indian Chapter) organized by IIT Madras and University of Leeds from
9		Flood Management in sensitive catchments of India: An appraisal through Hydrodynamic inundation modeling	30 November 2024 NIT Rourkela River Basin Management: Theory & Practice" workshop
10		Invited talk on "Compound Risks from Floods and Droughts through a Novel Concept of Bivariate Risk Classifier"	23-26 October 2024 Barcelona, Spain Multi-hazard DRR Academy for young researchers and
11		A Socio-Technocratic approach towards effective Flood Management Status and Future Challenges	20 to 24 May 2024 NIT Patna Recent Advances in Water Resources Engineering and Applications (RAWREA - 2024), organized by the Department of Civil Engineering, NIT Patna
12		Invited lectures on "Urban Flood Risk Mapping and Flood Plain Zoning"	1 to 4 July, 2024 Central Water Commission (CWC), Pune Training Programme on "Urban Flood Management" organized by the National Water Academy
13	2024	Invited lecture on "Preparedness and evacuation during emergency periods"	16 to 19 April CEC IIT Roorkee Training Programme on "Water Built Environment & Emergency Preparedness" under the aegis of National Health Mission, Odisha
14		Invited lecture on "Preparedness and evacuation during emergency periods"	29 April to 2 May CEC IIT Roorkee Training Programme on "Water Built Environment & Emergency Preparedness" under the aegis of National Health Mission, Odisha
15		Invited lecture on "Preparedness and evacuation during emergency periods"	6 to 9 May CEC IIT Roorkee Training Programme on "Water Built Environment & Emergency Preparedness" under the aegis of National Health Mission, Odisha
16		Invited lecture on "Preparedness and evacuation during emergency periods"	13 to 16 May 2024 CEC IIT Roorkee Training Programme on "Water Built Environment & Emergency Preparedness" under the aegis of National Health Mission, Odisha
17		Invited lecture on "Climate resilience and interventions- Engineering design, planning and implementation"	21 March 2024 Jaipur (India) Indian Institute for Human settlements (IIHS) under the aegis of Jal Jeevan Mission (JJM)

S No.	Year	Title of Lecture/ Lecture Series	Date, Place and Programme
18		Invited lecture on “Impact of climate change on water resources”	1 to 5 February, 2024 CEC, IIT Roorkee Training programme on Hydrological Data Collection, Processing, and Analysis for Surface Water Projects
20		Invited lecture on “Processing and analysis of precipitation data”	1 to 5 February, 2024 CEC, IIT Roorkee Training programme on Hydrological Data Collection, Processing, and Analysis for Surface Water Projects
21		Invited for a hands-on on “Processing and analysis of precipitation data”	1 to 5 February, 2024 CEC, IIT Roorkee Training programme on Hydrological Data Collection, Processing, and Analysis for Surface Water Projects
22	2023	Invited lecture on “Flood Risk Management”	21 October 2023 Iranian Hydraulic Association
23		Invited lecture on “Sustainable Flood management: Impacts of Climate change and Socio-economic dynamics”	12 October 2023 JNU, New Delhi One-Day workshop on Flood Modelling: Real-time Forecasting, Warning and Dissemination
24		Invited lecture on Adaptation to Climate Change on the “Climate Change Impacts Assessment and its Responses”	07 March 2023 NEERI Nagpur 2-day Skill Training Program on Climate Change Impacts Assessment and its Responses
25		Invited lecture on Applications of RS and GIS for water resources management	26 February 2023 Advanced Entrepreneurship and skill development programme (ESDP) on Irrigation Water Management Using Soil Moisture and Canal Discharge Data
26		Lecture on “Hydrodynamic Flood Inundation and Hazard Modeling: Current approaches and future challenges”	13 September 2023 Capacity Building Programme on Recent Advances in Satellite-based Technologies for monitoring Hydro-climatological ExtRemes (RASTER 2023) WRD&M, IIT Roorkee
27		Hands-on session on “Developing an exhaustive 1D-2D coupled flood model set-up using MIKE+”	14 September 2023 Capacity Building Programme on Recent Advances in Satellite-based Technologies for monitoring Hydro-climatological ExtRemes (RASTER 2023) WRD&M, IIT Roorkee
28		Lecture on “Capturing Flood Risks over coastal multi-hazard catchments by blending flood hazards and vulnerabilities”	14 September 2023 Capacity Building Programme on Recent Advances in Satellite-based Technologies for monitoring Hydro-climatological ExtRemes (RASTER 2023) WRD&M, IIT Roorkee
29	2022	An invited lecture on “Flood Forecasting Models”	06 December 2022 WRD&M, IIT Roorkee Training program on “Latest Technologies in Water Sector
30		An invited lecture on “Shifting towards a Socio-technocratic Approach for Sustainable Flood Risk Management”	06- 08 November 2022 INSA, New Delhi ACS Environment and Sustainability event
31		Hands-on-session on Hydraulic Modelling using TUFLOW	28 and 29 October 2022 WRD&M, IIT Roorkee

S No.	Year	Title of Lecture/ Lecture Series	Date, Place and Programme
32		One-day workshop on “Hydraulic modeling using TUFLOW”	21 October 2022 (online) WRD&M, IIT Roorkee
33	2021	An invited lecture on “Water management”	August 5, 2021 (online) IIM Vishakhapatnam
34		Invited lecture on “Fidelity of reanalysis datasets in floodplain mapping: Investigating performance at inundation level over large regions”	24 July 2021 in Lima, Peru (online) CONECTA AL II, CIVGEO 2022
35	2018	Invited lecture on “Advances in Urban Water Management”	10 to 14 Dec. 2018 CUSE, IIT Bombay
36		Invited lecture for a workshop on “Flooding in Cities, Warning & Dissemination”	20 and 21 August 2018 SVNIT, Surat, Gujarat (India)
37	2016	Invited lecture on Flood Modeling: A demonstration	30 May to 3 June 2016 CEP-QIP course on Impacts of Climate Change, Urbanization, and Land-Use-Land-Cover Change on Water Resources, IIT Bombay

TEACHING EXPERIENCE

Year and Semester	Course (number and name)	Credits (L-T-P)	Co-instructors (if any)	Remarks	Score	
					FS	CS
2022-23	WRN-571: Design of Irrigation Structures and Drainage Works	4 (3-1-0)	Prof. S. K Mishra	50% sharing	4.44	4.68
	WRN-517: River Engineering	4 (3-1-0)	--	--	4.91	4.92
	WRN-505: Drinking-Water and Sanitation Sustainability	4 (3-1-0)	--	--	4.90	4.94
2023-24	WRN-571: Design of Irrigation Structures and Drainage Works	4 (3-1-0)	Prof. S. K Mishra	50% sharing	4.59	4.70
	WRL-507: River Engineering	4 (3-1-0)	--	--	4.81	4.76
	WRC-543: Drinking-Water and Sanitation Sustainability	4 (3-1-0)	--	--	4.59	4.70
2024-25	WRN-571: Design of Irrigation Structures and Drainage Works	4 (3-1-0)	Prof. S. K Mishra	50% sharing	4.72	4.73
	WRL-507: River Engineering	4 (3-1-0)	--	--	4.80	4.77
	WRC-543: Drinking-Water and Sanitation Sustainability	4 (3-1-0)	--	--	4.85	4.92
2025-26	WRC-543: Drinking-Water and Sanitation Sustainability	4 (3-1-0)	--	--	4.70	4.73
	WRS-502: Policy and Strategies for DRR, Adaptation, Safety, and Sustainability	2 (2-0-0)	--	--	4.68	4.66

*L: Lecture, T: Theory, P: Practical; FS: Faculty score (out of 5), CS: Course Score (out of 5)

RESEARCH SUPERVISION

Ph.D. Supervision

S.No.	Name of Student	Completed/ Ongoing	Thesis Title	Other supervisor (s) (if any)
1	Dev Anand Thakur [Visiting DAAD fellow during PhD, and currently Postdoc Fellow at the University of Alabama, Tuscaloosa, United States]	2021, FT [Awarded the “Excellence in Doctoral Research Award” 2025 by IIT Roorkee]	A Unified Framework of Multi-Hazard Flood Risk Mapping under Changing Climate: An Application over a Coastal Catchment in India	NIL
2	Hrishikesh Singh [Prime Minister’s Research Fellow]	2022, FT	Unraveling long-term flood risk dynamics over India: Lessons for developing adaptation and resilience strategies	NIL
3	Vaibhav Tripathi	2023, FT	Leveraging Geomorphic Classifiers and Machine Learning Techniques for characterizing Flood Resilience	NIL
4	Rahul Deopa	2023, FT	Quantifying Climate-sensitive Hydrological and Public-Health Risks from Urban Flooding: Evidence from Hydrodynamic-cum-Water Quality Modeling and Vulnerability Assessment	NIL
5	Mayank Tyagi	2023, FT	Compound Risks from Floods and Droughts: Deciphering Hazard Inter-dependencies and Interactions under Changing Climate	NIL
6	Susmita Saha	2024, FT	Understanding Glacial Lake Formation and Outburst Flood Propagation under Climatic and Anthropogenic Stressors in the Indian Himalayas	Prof. Ashish Pandey
7	Rachit	2024, FT	Quantifying Climate- and Weather-Induced Multi-Hazards Using Deep Learning Models across the Indian Subcontinent	Prof. Ashish Pandey Prof. Anil K. Gupta
8	Munish Kumar	2024, FT	Water Security under Climate and Human Stressors: Modeling Water Availability and Allocation in a Sensitive Catchment of India	NIL
9	Ankush Charavande	2025, FT	Urban Flood Modeling and Resilience Assessment through Digital Twins: A Multi-Platform Approach for Indian MegaCities	NIL
10	Debasish Mishra	2025, PT	Understanding Hydroclimatic Extremes in Reservoir-Regulated Catchments: Spatio-Temporal Dynamics of Floods and Droughts	NIL
11	Dilip Barman	2022, PT	Identification of critical soil erosion-prone areas and preparation of the CAT Plan	Main Supervisor: Prof. Ashish Pandey
12	Brook Legese	2023, FT	Integrated Assessment of Climate Change and Land Use/Land Cover Dynamics on Water Resources in the Ziway Abijata Basin, Ethiopia	Main Supervisor: Prof. Ashish Pandey

S.No.	Name of Student	Completed/ Ongoing	Thesis Title	Other supervisor (s) (if any)
13	Deepak Pandey	2025, FT	Assessing the Impacts of Climate Change on Human Health: Pathways, Risks, and Adaptation in India	Main Supervisor: Prof. Anil K. Gupta Prof. Ashish Pandey

M.Tech. Supervision

Sl. No.	Batch	Names of Student	Title of Project/Thesis	Name of other supervisor (if any)
1	2021-22	Ms. Sakshi Tantway	Water Scarcity Assessment and Mitigation in Tikamgarh District of Bundelkhand Region	Main Supervisor: Prof. Deepak Khare
2		Mr. Vivek Kumar Meena	Storm Water Management in Urban areas using Remote Sensing and modelling using SWMM technology	Main Supervisor: Prof. Ashish Pandey
3	2022-23	Mr. Trashi Namgyal	An Integrated Framework to Quantify Flood Risks and Decoding Impacts on Interdependent Infrastructure Systems over Flood-Prone Mountainous Landscapes	NIL
4		Ms. Vishveshwari Tukaram Shirsath	Characterization of Flood Risk induced by Nuisance Flooding: A Demonstration over Mithi River Catchment, Mumbai	NIL
5		Mr. Rajesh Singh	Performance Evaluation of Multiple State-of-the-art Satellite Products for Precipitation Estimation and Flood-causing Potential over Nepal	NIL
6		Mr. Naresh Sharma	A Comprehensive Approach for Glacial Lake Outburst Flood Risk Assessment over Nepal	NIL
7		Ms. Punyo Asha	A Comprehensive Framework to Quantify the Impacts of Climate Change on Human Health Risks Emerging from Contaminated Urban Floodwaters	NIL
8		Mr. Gaurav Karna	Investigating the Arrangement and Inclination of Submerged Vanes for Controlling Local Scour under Various Hydraulic Conditions	NIL
9	2023-24	Mr. Gufran Alam	Demarcating Flood Risk Zones by Considering Geomorphic Classifiers through an Advanced Machine Learning-cum-SHAP framework	NIL
10		Ms. Renuka Ashok Kumar Shinde	Characterizing Net Positive Water Impact through Life Cycle Assessment: An Index-Based Approach for Evaluating Wastewater Treatment Systems	NIL

Sl. No.	Batch	Names of Student	Title of Project/Thesis	Name of other supervisor (if any)
11		Mr. Aashutosh Kumar Hirnotia	An Integrated Framework of Reconstructing Fluvial Architecture for Hydraulic Modelling: Application over a Data-scarce Himalayan river in India	Prof. S.K. Mishra
12	2024-25	Mr. Aakash Choudhary	Quantifying Dynamics in Ecosystem Services under Land Use Changes: Implication on Water Demand and Supply	NIL
13		Ms. Aditi Singh	A Comprehensive Assessment of Water Management through WSC Index: An Application over Mega Urban Centres in India	NIL
14		Ms. Nandana P.S.	Human-Health Risks from Urban Flooding: Integrating Remote Sensing Observations and Hydrodynamic-cum-Water Quality Modeling	NIL
15		Mr. Debasish Mishra	Hydrological Impact Assessment of Future Climate Change on a large River Basin in India	NIL
16		Mr. Purshottam Sharma	Improving Streamflow Prediction with LSTM networks: Demonstration over a data-scarce Flood-prone Basin in India	NIL
17		Ms. Priyanka Tripathy	Quantifying Composite Vulnerability to Flood and Droughts: A Nation-wide assessment by Integrating TOPSIS and Shannon Entropy techniques	NIL
18		Ms. Vaishnavi Bhagawan Patil	Effectiveness of Spur Dikes for preventing Erosion in Meandering Channels: Laboratory and Numerical Investigations	NIL
19		Ms. Kankanam Pathiranage Ayesha Dewmini	Efficacy of Hybrid River Training Works for Alleviating Bank Erosion	NIL
20		Mr. Sumit Bhatt	Understanding and modelling extreme multi-hazard events	NIL
21	2025-26	Mr. Charan	Development of a ViNGe (Virtual Man-made Drainage Network Generation" tool) for urban flood modelling over data-scarce regions	NIL
22		Mr. Pushparaj Singh	Assessment of Stormwater Drainage Systems through Hydrological and Water quality modelling	NIL
23		Mr. Govind Sah	Water distribution system modelling: applications through remote sensing and hydraulic modelling	NIL
24		Mr. Mukesh Singireddy	Quantifying sensitivity of topographic data in Glacial Lake Outburst Flood (GLOF) Risk Assessment	NIL

Sl. No.	Batch	Names of Student	Title of Project/Thesis	Name of other supervisor (if any)
25	2026-27	Mr. Cabangani Dube	Impacts of Climate Change on Agricultural Water Management	NIL
26		Ms. Happy Ogendi Mugondo	Quantification of climate induced shifts in stream flow regimes	NIL
27		Mr. Sanjeev Thilakasiri Abeysingha Jayawardana Yapa	Spatiotemporal dynamics of saltwater intrusion in the coastal zone: Application of remote sensing and GIS	NIL
28		Ms. Pedma Angmo	Impact of settlement growth and tourism-driven urbanization on drinking water sustainability in cold mountainous regions	NIL
29		Mr. Krishna Prasad J S	Integrated water management For climate-resilient cities	NIL
30		Mr. Aabbci J S Roodia	Life Cycle Environmental Assessment of Waste-to-Energy Technologies in India	NIL
31		Mr. Obvious Mhete		NIL

OUTREACH PROGRAMMES- CONFERENCES, WORKSHOPS, AND CAPACITY BUILDING

Name of the Conf./Seminar/Course	Sponsored By	Dates
Water, Built Environment & Emergency Preparedness	National Health Mission, Odisha	13 May to 16 May 2024
Water, Built Environment & Emergency Preparedness	National Health Mission, Odisha	06 May to 09 May 2024
Water, Built Environment & Emergency Preparedness	National Health Mission, Odisha	29 April to 02 May 2024
Water, Built Environment & Emergency Preparedness	National Health Mission, Odisha	11 April to 15 April 2024
The International Conference on the Future of Water Resources (ICFWR-2024)	Indian Water Resources Society (IWRS), Central Water Commission, Ministry of Jal Shakti	18 January to 20 January 2024
Recent Advances in Satellite Technologies for monitoring Hydro-climatological Extremes (RASTER 2023)	Indian Space Research Organization (ISRO)	11 Sept. to 15 Sept., 2023
e-ITEC course on “River Engineering and Management”	ITEC Programme, Development Partnership Administration (Ministry of External Affairs, Government of India)	13 to 15 Mar. 2023
ACS Editor's Workshop	American Chemical Society	10 Nov. 2023

Name of the Conf./Seminar/Course	Sponsored By	Dates
International Conference on Climate and Weather-related Extremes (ICCWE 2022)	IIT Roorkee, and SERB-DST	19 and 20 September 2022
Hands-on session on Hydraulic Modelling using TUFLOW	WRD&M, IIT Roorkee	28 and 29 October 2022
One-day workshop on “Hydraulic modeling using TUFLOW”	WRD&M, IIT Roorkee	21 October 2022 (online)
e-ITEC course on “Application of Remote Sensing and GIS in Agriculture”	ITEC Programme, Development Partnership Administration (Ministry of External Affairs, Government of India)	20-22 July 2022

ADMINISTRATIVE CONTRIBUTIONS

Department Level

- Prof I/C of River Engineering Lab [2021-present]
- Faculty Advisor (for Ph.D. research scholars) [2022-present]
- Prof I/C, Class, and timetable [2022-present]
- Member Secretary, Departmental Faculty Committee (DFC) [2022-2024]
- Member, Departmental Faculty Search Committee (DFSC) [2022-2025]
- Departmental Research Committee (DRC) [2021-present]
- Member, Department Academic Programme Committee (DAPC)

Institute level

- Warden, Ravindra Bhawan, IIT Roorkee
- Mentor, Bureau of Indian Standards (BIS, Student Chapter at WRD&M)
- Member, Standing Committee of Technical Experts for the stormwater management plan at IITR
- Member, Committee for Non-Convocation awards at IITR
- Representative from the department to the Dean of Resources and Alumni Affairs (DORA) advisory committee [2021-23]

COMPUTATIONAL SKILLS

- Hydraulic/hydrologic packages: MIKE 11, MIKE 21, MIKE URBAN, MIKE FLOOD, MIKE SHE, CaMa-Flood, TUFLOW, LISFLOOD-FP, HEC-HMS, HEC-RAS, SWAT, HYPE
- GIS packages: ArcGIS and QGIS
- Programming packages: MATLAB, R, Python
- Statistical packages: SPSS
- Operating System: Windows, Linux

Updated as of 13 August 2026